

Maths Chapter 10 Linear Equation in two Variables

Subject: Maths
Grade: class 10
Duration: 55 min

Total Marks: 30

Total Questions: 18

Due: 13 Jun 2026

Name: _____ Roll Number: _____ Section: _____

General Instructions:

1. Read all questions carefully before attempting to answer.
2. Do not write anything on the question paper except your name and roll number.
3. Use a pencil to shade your answers in the MCQ section.
4. Time Limit 55 Minutes Only

SECTION A: MULTIPLE CHOICE QUESTIONS

[10 marks]

Choose the most appropriate answer from the given options.

- Q1.** What is the equation of a line that passes through the points (1,2) and (3,4)? Easy [1 mark]
- A. $2x - y - 0 = 0$ B. $x - y + 1 = 0$
C. $x + y - 3 = 0$ D. $x - 2y + 1 = 0$
- Q2.** Solve for y: $2x + 3y = 7$ Medium [1 mark]
- A. $y = (7 - 2x) / 3$ B. $y = (7 + 2x) / 3$
C. $y = (7 - 2x) / 2$ D. $y = (7 + 2x) / 2$
- Q3.** The graph of the equation $2x + 5y = 10$ is a ____ Easy [1 mark]
- A. straight line B. circle
C. parabola D. ellipse
- Q4.** The equation of the line passing through the point (2,3) and having a slope of 2 is Medium [1 mark]
- A. $y - 3 = 2(x - 2)$ B. $y - 3 = 2(x + 2)$
C. $y - 2 = 3(x - 3)$ D. $y - 2 = 3(x + 3)$
- Q5.** If the equation $5x + 2y = 10$ represents a line, then the slope of this line is ____ Hard [1 mark]
- A. $-5/2$ B. $-2/5$
C. $5/2$ D. $2/5$
- Q6.** The equation of a line passing through (1,2) and parallel to the line $3x - 4y = 10$ is Hard [1 mark]
- A. $3x - 4y - 5 = 0$ B. $3x - 4y + 5 = 0$
C. $4x - 3y - 5 = 0$ D. $4x - 3y + 5 = 0$
- Q7.** Solve for x: $x/4 + 2 = 9$ Easy [1 mark]
- A. $x = 28$ B. $x = 29$
C. $x = 30$ D. $x = 31$
- Q8.** The equation of a line with slope 3 and y-intercept 2 is Medium [1 mark]
- A. $y = 3x + 2$ B. $y = 3x - 2$
C. $y = 2x + 3$ D. $y = 2x - 3$

- Q9.** Solve for y: $y - 2 = 3(x - 1)$ Medium [1 mark]
- A. $y = 3x - 1$ B. $y = 3x + 1$
 C. $y = 2x + 1$ D. $y = 2x - 1$
- Q10.** The equation of the line passing through the points (0,0) and (2,4) is Hard [1 mark]
- A. $x - y = 0$ B. $x + y = 0$
 C. $2x - y = 0$ D. $x + 2y = 0$

SECTION B: SHORT ANSWER QUESTIONS

[8 marks]

Answer all questions briefly and to the point.

- Q11.** Solve for x and y: $2x + 3y = 12$ and $x - 2y = -3$ Easy [2 marks]
- Answer: _____
- Q12.** Find the equation of the line passing through the point (2,3) and having a slope of -2. Medium [2 marks]
- Answer: _____
- Q13.** The sum of the digits of a two-digit number is 9. If the tens digit is x and the units digit is y, write an equation that represents this situation. Hard [2 marks]
- Answer: _____
- Q14.** Solve for x: $x/2 + 5 = 11$ Medium [2 marks]
- Answer: _____

SECTION C: LONG ANSWER QUESTIONS

[12 marks]

Answer all questions in detail. Use examples where appropriate.

- Q15.** Solve the system of equations: $x + 2y = 7$ and $2x - 3y = -3$. Show all steps and explain your reasoning. Hard [3 marks]
- _____
- _____
- _____
- _____
- Q16.** Find the equation of the line passing through the points (1,2) and (3,4). Explain how you found the slope and y-intercept. Medium [3 marks]
- _____
- _____
- _____
- _____
- Q17.** The cost of a ticket to a movie is \$5 for adults and \$3 for children. If 200 people attend the movie and the total revenue is \$800, write an equation that represents this situation and solve for the number of adults and children. Hard [3 marks]
- _____
- _____
- _____
- _____

Q18. Graph the equation $2x + 3y = 6$ and find the x and y intercepts. Explain how you found the intercepts and what they represent.

Medium

[3 marks]
